

Re-integrating Prosit (and PTM) Koina Models into Pioneer BuildSpecLib

A phased plan toward phosphoproteomics support

Pioneer.jl — prepared for N. Wamsley

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1. Executive summary

Pioneer historically supported several Koina fragment-intensity models (Prosit, UniSpec, AlphaPept-Deep) alongside Altimeter and Chronologer. That multi-model support was removed on **2026-05-01** (commit 6bab0dadb, “*Consolidate SearchDIA pipeline*”) to simplify the codebase down to a single fragment model (Altimeter, a `SplineCoefficientModel`) and a single RT model (Chronologer, a `RetentionTimeModel`).

We now want Prosit back — specifically because Koina hosts **Prosit models trained with post-translational modifications (PTMs)**, opening a path to phosphoproteomics. This document is the plan.

What I verified before writing this plan (not assumptions):

- This environment **can reach and run inference on** `koina.wilhelmlab.org`. I ran real requests, not just health checks.
- The PTM intensity model `Prosit_2024_intensity_PTMs_gl` works: I sent `S[UNIMOD:21]PEPTIDEK` (phospho-Ser) and the b2 ion correctly shifted +79.966 Da.
- A **matched iRT model** `Prosit_2024_irt_PTMs_gl` exists and returns iRT for the same PTM peptides.
- **Chronologer already accepts PTMs** — I sent phospho (UNIMOD:21), oxidation (UNIMOD:35), and carbamidomethyl (UNIMOD:4) and it returned shifted RTs. This means for the phospho case we have **two** viable RT paths (reuse Chronologer, or Prosit’s matched iRT).
- The old Prosit code and a **working Prosit build config** still exist in git history and on disk (`data/test_build_spec_lib/scenario_prosit/params_prosit.json`, pointing at `minimal_protein.fasta` — a single-protein FASTA ideal for the first test).

The three real engineering problems (all confirmed against the code and the API):

1. **Raw scalar intensities, not splines.** Altimeter returns NCE-dependent B-spline *coefficients*; Prosit returns a single fixed intensity vector. Different storage type, different search-time evaluation.
2. **No NCE calibration.** Altimeter’s spline *is* the calibration (evaluated at a per-run NCE at search time). Prosit takes a raw `collision_energies` input and needs the old static-NCE + charge-adjustment machinery restored.

3. **Monoisotopic vs total-abundance fragment intensities.** This is the subtle one. Pioneer's library stores *total* fragment abundance across the isotope envelope and redistributes it at search time by sulfur count (`getFragIsotopes!`). Prosit predicts only the **monoiso-topic** peak. Because the monoisotopic fraction of total abundance varies with fragment mass/composition, feeding Prosit's number in directly is *inconsistent across fragments*. We need a defined conversion.

The plan below re-adds Prosit surgically (the removal commit is **not** cleanly revertible — it bundled unrelated refactors), validates against Altimeter/Chronologer on non-PTM data first, and only then moves to phosphoproteomics.

2. Goals, non-goals, and phasing

2.1 Definition of done for Phase 1 (this plan's primary target)

Pioneer can build a spectral library using a Prosit Koina model for fragment intensities, on unmodified peptides, and a DIA search using that library performs **comparably to** the Altimeter/Chronologer library on the same real data.

"Comparably" = within a small, agreed tolerance on precursor/protein IDs at 1% FDR **and comparable quant** on the **MTAC yeast 3M run** (yeast = fastest full library to build; we set the exact tolerance once we see the reference Altimeter numbers — see §8).

2.2 Phase map

Phase	Outcome	Gate to next phase
0. Scaffolding	Prosit model dispatch re-added; one-protein library builds end to end via Koina.	Library file is produced; fragments look sane vs a hand-checked Koina response.
1. Non-PTM parity	Prosit library searches real DIA data; results benchmarked vs Altimeter.	IDs within tolerance of Altimeter on the shared dataset.
2. PTM plumbing	Prosit_2024_intensity_PTMs_g(+ RT) build phospho libraries; UNIMOD encoding verified.	Phospho library builds; fragment m/z shifts verified for known peptides.
3. Phospho search + localization	Search a phospho DIA dataset; add/verify site localization.	Runs on a benchmark phospho dataset with sane localization.
4. Scale	Handle combinatorial PTM library size (indexing, memory, decoys).	Large phospho library builds and searches within resource budget.

Phases 2–4 are **future work** scoped here only enough to make sure Phase 1 choices don't paint us into a corner. Phase 1 is where the near-term effort goes.

2.3 Non-goals (explicitly out of scope for now)

- Re-adding UniSpec and AlphaPeptDeep. The old dispatch made them cheap to add later, and we will preserve that generality, but we will not test or maintain them now.
- Training or fine-tuning any model. We consume Koina as-is.
- Site-localization algorithm design (Phase 3) beyond noting the integration point.

3. Current architecture (what exists today)

Everything dispatches on a concrete `KoinaModelType` subtype (`src/structs/KoinaStructs/KoinaModelTypes.jl`). Today only two remain:

```
abstract type KoinaModelType end
struct SplineCoefficientModel <: KoinaModelType; name::String end # Altimeter (fragments)
struct RetentionTimeModel <: KoinaModelType; name::String end # Chronologer (RT)
```

The removed ones were `InstrumentSpecificModel` (`UniSpec`, `AlphaPeptDeep`) and `InstrumentAgnosticModel` (`Prosit`). **Prosit was an `InstrumentAgnosticModel`** — it ignores instrument type but does take a collision energy.

Build pipeline (all under `src/Routines/BuildSpecLib/`, entry `src/Routines/BuildSpecLib.jl`):

```
FASTA digest + mods + decoys      chronologer/chronologer_prep.jl
  -> precursor table (koina_sequence, charge, collision_energy, mz, sulfur)
RT prediction (Chronologer/Koina)  chronologer/chronologer_predict.jl -> precursors.arrow
Fragment prediction (Altimeter)    fragments/fragment_predict.jl -> raw_fragments.arrow
  prepare_koina_batch / parse_koina_batch  koina/koina_batch_prep.jl, koina_batch_parse.jl
  filter_fragments! (::SplineCoefficientModel, ctx) (predict-time filtering)
Decode -> library fragment structs  fragments/fragment_parse.jl (build_detailed_frags_from_raw)
Final indexes + serialize          build/build_poin_lib.jl (buildPionLib)
```

Model config lives in `src/Pioneer.jl`:

```
const MODEL_CONFIGS = Dict(
  "altimeter" => (annotation_type=UniSpecFragAnnotation("y1~1"),
                  model_type=SplineCoefficientModel("altimeter"), instruments=Set{String}{}))
const KOINA_URLS = Dict(
  "chronologer" => ".../v2/models/Chronologer_RT/infer",
  "altimeter"   => ".../v2/models/Altimeter_2024_splines_index/infer")
```

`BuildSpecLib.jl` currently **hardcodes** `prediction_model = "altimeter"` and the config schema no longer accepts `prediction_model/instrument_type` (`utils/check_params.jl`:116–118 documents this).

Two representations of a fragment already exist in `src/structs/SpectralLibrary/fragment_types.jl`:

- `SplineCompactFrag{N,T}` — carries an `NTuple{N,T}` of spline coefficients; `getIntensity(::SplineCompactFrag, ::SplineType)` evaluates the spline at NCE at **search time**. (Altimeter.)
- `DetailedFrag{T}` — carries a scalar intensity `::Float16`; `getIntensity(::DetailedFrag, ::ConstantType)` returns it directly. **Currently unused by the build — this is the natural target for a raw-intensity Prosit fragment.**

So the scalar-intensity path is *scaffolded but disconnected*. That is the single most important fact for scoping: we are re-wiring an existing shape, not inventing one.

4. Koina capabilities (empirically verified 2026-07-02)

Server is a Triton inference server. Useful endpoints:

- GET /v2/models/{model}/config — I/O contract
- POST /v2/repository/index — full model list
- POST /v2/models/{model}/infer — inference({{"id","inputs":[{"name,shape,datatype,data}]}})

4.1 Models relevant to us (all confirmed present)

Purpose	Model	Notes
Non-PTM intensity	Prosit_2020_intensity_HCD (_CID,_TMT)	Phase 1 workhorse
PTM intensity	Prosit_2024_intensity_PTMs_g1	Phase 2; extra input fragmentation_types
PTM intensity (newer)	Prosit_2025_intensity_22PTM, _40PTM	more PTM classes
PTM iRT (matched)	Prosit_2024_irt_PTMs_g1, Prosit_2025_irt_*	RT alternative to Chronologer
Current RT	Chronologer_RT	also accepts PTMs (verified)

4.2 I/O contracts (verified)

Prosit_2020_intensity_HCD

```
IN : peptide_sequences STRING[-1], precursor_charges INT32[1], collision_energies FP32[1]
OUT: intensities FP32[174], mz FP32[174], annotation STRING[174] # fixed 174 slots
```

Prosit_2024_intensity_PTMs_g1 # NOTE the 4th input

```
IN : peptide_sequences, precursor_charges, collision_energies, fragmentation_types STRING[1]
OUT: intensities/mz/annotation [174]
```

Prosit_2024_irt_PTMs_g1 : IN peptide_sequences -> OUT irt FP32[1]

Chronologer_RT : IN peptide_sequences -> OUT rt FP32[1]

Contrast — Altimeter is structurally different (this is why it needs its own path and Prosit needs the *other* path):

```
Altimeter_2024_splines : OUT coefficients FP32[4,200], knots FP32[8] # NCE-dependent spline
Altimeter_2024_reisotope : IN fragment_masses, precursor_sulfurs, fragment_sulfurs,
                           isotope_isolation_efficiencies[5] -> OUT intensities[1900] # 380
```

Altimeter models the **full isotope envelope internally** (sulfur-aware, isolation-efficiency aware) and returns *total* abundance as a spline. Prosit returns a single monoisotopic scalar at one CE. That difference is §5.3.

4.3 Verified inference (phospho round-trip)

Prosit_2024_intensity_PTMs_g1, charge 2, CE 27, HCD:

- PEPTIDEK -> 28 non-zero peaks, base peak y6.
- S[UNIMOD:21]PEPTIDEK -> b2 227.103 -> **265.058** (+79.966, phospho on Ser1). Correct.
- Matched Prosit_2024_irt_PTMs_g1 -> iRT [4.06, 17.59, 108.12].

Intensities are normalized to base peak = 1.0. Slots that don't apply come back -1 / 0 and must be filtered (there are always 174 slots).

5. The three divergences and how we handle each

5.1 Raw intensities vs spline coefficients

Fact. Prosit returns intensities (scalar per fragment); Altimeter returns coefficients + knots.

Plan. Prosit fragments become DetailedFrag{Float16} (scalar) with ConstantType intensity dispatch — the *already-present* unused path. No spline knots file (spline_knots.jls), no SplineFragmentLookup. The build step (build/build_poin_lib.jl) gets a buildPionLib(::InstrumentAgnosti...) method that mirrors the spline one but uses the constant-intensity lookup. The old code did exactly this (process_batch! vs process_spline_batch!), so we have a template.

5.2 NCE calibration

Fact. Altimeter doesn't consume NCE at build time (spline evaluated per-run at search). Prosit needs a collision_energies value per precursor. The old static-NCE machinery was removed: adjustNCE(NCE, default_charge, peptide_charge, charge_fac) and CHARGE_ADJUSTMENT_FACTORS = [1, 0.9, 0.85, 0.8, 0.75], plus nce_params.default_charge and nce_params.dynamic_nce config fields.

Decision (2026-07-02): flat NCE, no per-charge adjustment — 0.4 deferred. This is already the current behavior: chronologer_prep.jl populates :collision_energy from nce_params.nce (a flat 26.0), and Prosit's prepare_koina_batch sends that as collision_energies. So the Prosit build already works with a flat NCE and **no code change is needed**. Per-charge adjustment (adjustNCE + CHARGE_ADJUSTMENT_FACTORS + nce_params.default_charge/dynamic_nce) is an *optimization*, not correctness-critical, and is explicitly deferred — we can add it later if the benchmark shows it matters. (UniSpec's per-run m/z->eV calibration is a separate mechanism we are also not restoring — out of scope.)

Still to tune in Phase 1: the flat NCE *value*. Prosit intensity is CE-sensitive, so a wrong global NCE hurts spectral-contrast scores. We sweep a small NCE grid on the benchmark (the mirror-plot tool already shows Altimeter and Prosit NCE scales differ) and pick the best flat value — a scalar choice, distinct from the deferred per-charge adjustment.

5.3 Monoisotopic vs total-abundance intensities (the crux)

Fact. Pioneer stores *total* fragment abundance and, at search time, getFragIsotopes! (src/Routines/SearchDIA/CommonSearchUtils/getFragIsotopes.jl) splits that total across isotope peaks using iso_splines keyed on the fragment's sulfur count. Prosit predicts the **monoisotopic** peak only. The monoisotopic fraction of a fragment's total abundance depends on

its mass and composition (more mass -> smaller monoisotopic fraction), so Prosit's raw number is *not* on the same scale across fragments as Pioneer's total-abundance convention.

Two candidate strategies:

- **(A) Convert monoisotopic -> total at build time (recommended).** For each Prosit fragment, divide the predicted monoisotopic intensity by the expected monoisotopic isotope fraction for that fragment's mass and sulfur count (the same `iso_splines` model Pioneer already owns). The stored value is then a *total abundance* consistent with the Altimeter convention, and **the entire search-side isotope path is reused unchanged**. This is the cleanest and keeps one search code path.
- **(B) Store monoisotopic directly, special-case search.** Add a per-library flag that tells `getFragIsotopes!` the stored intensity is already monoisotopic and to build the envelope up from it (multiply by the envelope) rather than redistribute. More search-side surface area, but avoids a build-time division that can amplify noise for high-mass, low-fraction fragments.

Decision (agreed): strategy (A). It saves the search-time work of a special isotope path, keeps the code much simpler, isolates the change to the build side, and leaves the hot search path identical — which also makes the Phase-1 Altimeter comparison cleaner (only the library differs, not the search code). We still validate (A) numerically in Phase 0 by comparing a converted-Prosit fragment's implied envelope against a matched Altimeter/theoretical envelope for a handful of peptides before trusting it at scale, and keep (B) documented only as a fallback if (A) proves numerically fragile for high-mass, low-fraction fragments.

5.3.1 Concrete mechanism (decided)

Same isotope model everywhere. The mono->total conversion uses the *identical* `IsotopeSplineModel` (`src/Utils/isotopeSplines.jl:85`) that search uses in `getFragIsotopes!` (`src/Routines/SearchDIA/CommonSe`

```
f0    = iso_splines(min(sulfur,5), 0, frag_mz*frag_charge)    # monoisotopic fraction
total = prosit_mono_intensity / f0                          # stored library intensity
```

This closes exactly: at search the reconstructed monoisotopic peak is `iso_splines(sulfur,0,mass)*total` = `P`, so Prosit's predicted peak is preserved and the `M+1/M+2...` are synthesized — the same treatment Altimeter's stored total receives. Using the same splines at build and search is what makes strategy A correct; a mass-only / average approximation is explicitly rejected.

Processing order (decided). Per precursor, within each predicted batch: metadata-filter -> **convert mono->total (splines) -> rank by total, keep topN** -> write. The convert must precede the rank because `f0` is mass-dependent (~1.3-2.9x across the fragment mass range) and reorders fragments.

Where it runs (decided). `raw_fragments.arrow` is written *already topN-filtered* today (`filter_fragments!` runs at `fragment_predict.jl:186`, before the Arrow write at :143-147; ~12% of predicted fragments survive). So the Prosit convert+rank+topN must run at **predict time** inside a `filter_fragments! (::InstrumentAgnosticModel)` overload — *not* at decode, or `raw_fragments.arrow` would carry all ~174 fragments/precursor and blow up.

What the predict-time path needs (new plumbing) — reuses existing functions. To evaluate the splines before the write, the filter context (`build_spline_frag_filter_ctx`) gains: (1) the loaded `IsotopeSplineModel` (load the isotope XML once), and (2) per-fragment **sulfur**, computed early from data we already have. The exact functions exist:

- `count_sulfurs!` (`fragment_parse.jl:199`) builds `seq_idx_to_sulfur` (1 per C/M residue – mod sulfur diffs at their positions) from (`plain_sequence`, `mods_iterator`, `mods_to_sulfur_diff`).
- `get_fragment_indices`(`base_type`, `index`, `seq_len`) (:236) gives a fragment's residue span (`b/a/c` -> (1,`index`); `y/x/z` -> (`seq_len`-`index`+1, `seq_len`); `p` -> whole).

`base_type/frag_index/charge` are already on the annotation the filter reads (`fragment_predict.jl:328-350`) and `prec_len` is on the `ctx`; the only new `ctx` input is the precursor **sequence + mods**. Plan: precompute a per-precursor **cumulative sulfur prefix array** when building the `ctx` (it already loops precursors at :256-259); then per fragment, `sulfur = prefix[stop] - prefix[start-1]` ($O(1)$).

Carry forward, don't recompute. For `Prosit`, `raw_fragments.arrow` stores the already-converted intensities (= total) *and* the per-fragment sulfur as columns; the decode step packs them straight into `DetailedFrag`. This guarantees identical sulfur/intensity from `predict` -> `decode` -> `search` (search reads `getSulfurCount(frag)` at `getFragIsotopes.jl:63`) with no double computation.

Rank filtering inventory (current, for reference). (1) `filter_fragments! (::SplineCoefficientModel)` = the only live per-precursor topN (order-based for `Altimeter`, since a spline has no scalar to sort by); (2) `getDetailedFrag/filterFrag` in `build_poin_lib.jl` = vestigial, “no longer consulted” (comment at :902-910); (3) the `SimpleFrag` fragment-index build (`build_poin_lib.jl:500-543`, `max_rank_index=8`) = a *separate* top-8 cap for the coarse search index, downstream of the detailed topN. `Prosit` changes only (1) — intensity-based instead of order-based.

6. Concrete attach points (surgical change list)

Each row is a place that currently has *only* a `SplineCoefficientModel` method and needs a parallel `Prosit` (`InstrumentAgnosticModel`) method. Old implementations are recoverable from `6bab0dadb^` (= `a33f96466`).

Concern	File : site	Prosit method to add
Model type	<code>structs/KoinaStructs/KoinaModelTypes.jl</code>	<code>add Prosit InstrumentAgnosticModel (and keep InstrumentSpecificModel stub for future)</code>
Registry	<code>src/Pioneer.jl</code> (<code>MODEL_CONFIGS</code> , <code>KOINA_URLS</code>)	<code>add prosit_2020_hcd (+ PTM entries later) w/ GenericFragAnnotation, URL</code>
Request	<code>koina/koina_batch_prep.jl:21</code>	<code>prepare_koina_batch (::InstrumentAgnosticModel) adds collision_energies (and fragmentation_types for PTM models)</code>
Parse	<code>koina/koina_batch_parse.jl:40</code>	<code>parse_koina_batch (::InstrumentAgnosticModel) reads scalar intensities+mz+string annotation</code>

Concern	File : site	Prosit method to add
Filter + rank	fragments/fragment_predict.jl:1298	filter_fragments! (::InstrumentAgnosticModel, metadata filters, then mono->total via iso_splines, rank by total, topN (§5.3.1). Ctx gains IsotopeSplineModel + per-fragment sulfur (needs precursor sequence on ctx).
Annotation	fragments/fragment_annotation.jl:237	GenieFragAnnotation parser already exists — reuse
Decode	fragments/fragment_parse.jl:681	Prosit DetailedFrag from intensities (template: old process_batch!)
Build	build/build_pion_lib.jl:132,869	BuildPionLib/getDetailedFrag (::InstrumentAgnosticModel, w/ ConstantType, no knots
NCE	chronologer/chronologer_prepare.jl:526	adjustNCE+CHARGE_ADJUSTMENT_FACTORS on Prosit branch
Isotopes	predict-time filter (§5.3.1)	mono->total using the same iso_splines search uses; total = P / iso_splines(min(sulfur,5),0,mass); search path unchanged
Config schema	utils/check_params.jl:116	re-add prediction_model, instrument_type, nce_params.default_charge
Wiring	BuildSpecLib.jl:159,279	branch on model type; skip knot metadata for Prosit

Note the removal commit 6bab0dadb deleted 2,584 lines across 20 files but *interleaved* unrelated work (diann_decoys, a new koina_client, mods refactor). **Do not revert it.** Use `git show 6bab0dadb^:<path>` as a *reference*, then hand-port onto today's code.

7. Implementation plan (Phase 0 -> 1), step by step

Each step states its verification, per CLAUDE.md §4.

- [DONE] 0.1 Re-add InstrumentAgnosticModel + registry entry (prosit_2020_hcd)
verify: MODEL_CONFIGS["prosit_2020_hcd"] resolves; precompiles. (commit 314886fdc)
- [DONE] 0.2 prepare_koina_batch + parse_koina_batch for Prosit
verify: request JSON matches §4.2; live parse yields intensities+mz+annotation, n_frgs=174, base-peak=1.0. (commit 1a46ab54d)
- [DONE] 0.3 filter (mono->total via iso_splines + sulfur + topN) + decode -> CompactFrag;
buildPionLib (::InstrumentAgnosticModel) -> StandardFragmentLookup. NOTE: emits CompactFrag (not DetailedFrag) - the modern build target; mono->total (0.5) folded into the predict-time filter; survivors reordered by total so rank=intensity rank.


```

verify: one-protein build (minimal_protein.fasta) -> 4 prec / 40 frags (10/prec);
       every ALYKMSR frag m/z matches live Koina, intensity==Float16(mono/f0) with
       correct sulfur, ranks intensity-ordered. Altimeter build regression passes.
       (commits 27fbf308d [0.3a filter], 59772198d [0.3b decode+build])
[DEFERRED] 0.4 Per-charge NCE adjustment - NOT done by decision (2026-07-02). Flat NCE
       (nce_params.nce=26) is the current behavior and needs no code change; per-charge
       adjustment is an optimization for later (§5.2). Phase 0 is complete without it.
[DONE] 1.1 Built full yeast prosit_2020_hcd + altimeter libs (355,188 prec, 10 frags/prec).
[DONE] 1.2 Searched 3 MTAC 3-min reps with each (default develop params, 8 threads, 1% FDR).
[DONE] 1.3 Benchmark: PROSIT ON PAR WITH ALTIMETER (see FINDINGS.md). Unique precursors
       51,237 vs 51,391 (-0.3%), 95% overlap, protein groups 4401 vs 4386 (~0%), quant
       median CV 16.1% vs 16.6% (Prosit marginally better). Phase-1 done-definition MET.
1.4 Small NCE sweep - OPTIONAL now: parity already achieved at flat NCE 27.
    verify: chosen NCE documented with the ID curve (if pursued).

```

We keep the **Altimeter path as the reference** throughout — same search code, same raw files, only the library differs.

7.1 Testing infrastructure to reuse

- One-protein fixture: `data/test_fasta_files/minimal_protein.fasta` (already referenced by the historical `params_prosit.json`).
- The Koina client abstraction (`koina/koina_client.jl`) provides a pluggable client selected per-task via `with_koina_client (:84-110)`. Four clients exist: `HttpKoinaClient` (production), `SyntheticKoinaClient` (deterministic offline fake — **Altimeter/Chronologer only**, throws on other endpoints, :265-280), `RecordingKoinaClient` (captures live (`request,url,response`) triples, :194-224), and `FixtureKoinaClient` (replays them, keyed by a canonicalized `request+URL` hash, errors on cache miss, :235-261). A recorded fixture already ships for the existing tests (`test/Routines/BuildSpecLib/koina_fixtures.json`).
- **How we use it for Prosit — with limits.** The `record->replay` path (`RecordingKoinaClient -> save_koina_fixtures -> FixtureKoinaClient`) is endpoint-agnostic, so it works for Prosit unmodified: record real Prosit responses once, then iterate on the *downstream* code (`parse -> mono->total conversion -> decode -> build`) against the fixture without re-hitting Koina. Caveats that bound this: (1) fixtures are keyed on the **exact** request payload, so changing the digest / charges / NCE / batch composition is a cache miss — replay speeds up downstream-code iteration with the request set held fixed, it is not a general “free rebuild”. (2) `SyntheticKoinaClient` does **not** cover Prosit as-is (Altimeter/ Chronologer only); for a network-free deterministic *unit* test we either use the fixture path or extend the synthetic client. (3) A full-yeast fixture would be large (the current small-scenario fixture is already ~17.5 MB; a full yeast library would be far bigger) — a local dev artifact, not something to commit. For CI we record a **small** Prosit scenario (one/few proteins) and commit that, keeping the live server a dev-only dependency.

8. Benchmark / real-data validation strategy

We validate at **two tiers**, cheapest first, so we catch problems before paying for a full search.

Tier 1 — library-level diff (fast, no search, no raw file). Build a single-protein library (`minimal_protein.fasta`) with `prosit_2020_hcd` and inspect the library itself: fragment m/z

vs a live Koina call, intensity ordering, the monoisotopic->total conversion (§5.3-A) against a theoretical/Altimeter envelope for a few peptides. This is the Phase-0 gate and needs nothing but the Koina server.

Mirror-plot sanity check (dev_docs/prosit_ptm_integration/mirror_prosit_vs_altimeter.jl, built in Phase 0.2): overlays Prosit (up) vs Altimeter (down, spline evaluated at NCE) predicted spectra on a shared m/z axis, swept over an NCE grid, each restricted to its **top 10 fragments by intensity** (matching the library's max_frag_rank topN) and self-normalized. First run (peptides LGGNEQVTR / GISNEGQNASIK / TASEFDSAIAQDK / LFLQFGAQGSPFLK, NCE {22,26,30}): fragment m/z **align** between the two models (core check passes), Altimeter fragments more with rising NCE, and the **NCE scales of the two models differ** (Altimeter looks under-fragmented at NCE 22 vs Prosit; base peaks don't always coincide) — an early, concrete confirmation that Phase-1 needs the NCE calibration/sweep (§5.2).

Tier 2 — full-pipeline parity (IDs + quant), kept deliberately simple. We need a real DIA run, but the *fastest full library to build* is **yeast** (small proteome), so the Phase-1 benchmark is the **MTAC yeast 3M run**. Even a full yeast Prosit prediction takes some time, but it is far cheaper than human/HeLa. Yeast MTAC raw data comes from **RIS** when we reach this step (not needed until Phase 1.2). Procedure:

1. Build a yeast library with Altimeter/Chronologer (reference) and one with prosit_2020_hcd (+ Chronologer RT).
2. Search the same MTAC yeast 3M raw file(s) with each, identical search params.
3. Compare **precursor and protein IDs at 1% FDR and quant** (e.g. quantified precursor/protein counts and CV/abundance agreement), Prosit vs Altimeter.

Metric detail: we also eyeball spectral-contrast/score distributions, not just counts, because a model swap can shift score scale without changing rank much. We agree the pass tolerance before running (e.g. “Prosit within X% of Altimeter precursor IDs and comparable quant CVs”) once we see the reference Altimeter numbers on this run.

9. Phosphoproteomics roadmap (Phases 2-4)

9.0 Phase 2 concrete plan (phospho, 2026-07-02)

Phospho = variable mod Unimod:21 (+79.966331) on S/T/Y (single UNIMOD id; residue gives specificity). Config: variable_mods name Unimod:21, pattern [STY], mass 79.966331.

Verified: - Koina PTM models all handle phospho with correct residue-specific mass placement, and agree tightly: Prosit_2024_intensity_PTMs_g1 vs 2025_22PTM cosine **1.000**, vs 2025_40PTM **0.998**. RT works via **Chronologer** (accepts phospho) and the matched Prosit iRT. Papers: 2024 = PROSPECT-PTMs (NeurIPS 2024); 2025 22/40PTM = Prosit-PTM data-augmented (bioRxiv 2025). For dev either is fine — both run without error. - **Variable-mod generation verified correct** (matchVarMods + fillVarModStrings!): LSSTPEYK (S2,S3,T4,Y7) yields 5 / 11 / 15 combinations at max_var_mods 1 / 2 / 3 (= Sum C(4,n)), correct positions/residues.

Model choice: default Prosit_2024_intensity_PTMs_g1 + Chronologer RT; 40PTM is a one-line drop-in (broader coverage, ~identical output) so we can use both without error.

Code changes (small): (1) register prosit_2024_ptm in MODEL_CONFIGS/KOINA_URLS with a fragmentation_type="HCD" field; (2) prepare_koina_batch(::InstrumentAgnosticModel) sends

fragmentation_types when the config specifies it. Everything else (parse, iso-spline mono->total filter, decode->CompactFrag, build->StandardFragmentLookup, Chronologer RT) is reused unchanged. Phospho adds no sulfur and P is monoisotopic, so mono->total is unaffected.

Gate: build the synthetic phospho library (synth_phospho.fasta, variable [STY] Unimod:21, max_var_mods=2) end to end; verify pS/pT/pY fragment m/z shifts and RT vs live Koina — the one-protein gate, with PTMs, on a debuggable fixture.

9.0.1 Variable-mod combinatorial limits (informed by Sage issue #231)

Sage [issue #231](#) (variable-mod limits) is the reference for how to bound combinatorial explosion in PTM-heavy samples. Pioneer today has **only** the global max_var_mods cap — no per-mod occurrence limit and no max_combinations per-peptide hard cap, and countVarModCombinations pre-allocates all_mods to the full binomial sum (a pathological peptide could OOM before any cap applies). **Decision:** correct as-is for phospho dev/testing at modest max_var_mods; at **scale (Phase 4)** add a max_combinations per-peptide cap (“prefer fewer-PTM forms”) + optional per-mod occurrence limit, guarding the pre-allocation, mirroring Sage #231’s design.

9.1 Longer-range (Phases 2-4, scoped)

Enough detail to avoid Phase-1 dead-ends:

- **Model:** Prosit_2024_intensity_PTMs_g1 for fragments. The PTM intensity model needs the extra fragmentation_types input, so the request-prep method must be PTM-aware.
- **RT (test both):** implement and compare **Chronologer** and **Prosit matched iRT** (Prosit_2024_irt_PTMs_g1). Chronologer is generally the more accurate RT predictor, so the default preference is Chronologer *when the requested PTM subset is within Chronologer’s supported modification alphabet* (verified to include phospho/ox/ carbamidomethyl; the full alphabet must be enumerated in Phase 2). Fall back to Prosit iRT for PTMs Chronologer cannot represent. We build the plumbing for both regardless.
- **UNIMOD encoding:** Pioneer’s getKoinaSeqs already emits [MODNAME] after residues; the historical format matches ProForma/UNIMOD. Verify the exact token Prosit expects ([UNIMOD:21]) matches what getKoinaSeqs emits — likely a small naming shim.
- **Site localization (Phase 3):** new algorithmic work. Integration point is post-search scoring; we will need a benchmark phospho dataset (e.g. a synthetic-phosphopeptide or well-characterized enriched dataset) to measure localization accuracy. Flagged as its own design doc when we get there.
- **Library scale (Phase 4):** variable phospho on S/T/Y is combinatorially explosive. Concerns: precursor-table size, fragment-index memory, decoy generation cost, and Koina request volume (batch of 1000, many batches). We will need to measure and probably cap site combinations per peptide. This is the biggest unknown and deserves its own plan.

10. Risks and open questions

1. **Isotope conversion (§5.3).** Strategy A vs B — the decision I most want your call on. Risk: build-time division amplifies error for high-mass low-fraction fragments.
2. **Default NCE (§5.2).** Wrong global NCE silently degrades Prosit spectral scores. Mitigated by the Phase-1 sweep.

3. **Non-clean removal commit.** Hand-porting from 6bab0dadb[^] risks missing a subtle coupling (e.g. the mods refactor changed since). Mitigated by building the one-protein library early (0.3) and diffing fragment output against live Koina.
4. **Prosit 174-slot ceiling.** Prosit caps peptide length / fragment count; very long peptides lose fragments. Altimeter uses 380/1900 slots. For typical tryptic peptides this is fine; flag if the benchmark has long peptides.
5. **CI dependence on Koina.** Mitigated by Recording/Fixture clients (§7.1).

Decisions (resolved 2026-07-02):

- (Q1) Isotope: **strategy A** — convert monoisotopic->total at build (simpler, cheaper). §5.3.
- (Q2) Benchmark: **two tiers** — single-protein library diff, then the **MTAC yeast 3M run** (fastest full library to build; data from RIS at Phase 1.2). Pass tolerance set once we see the reference Altimeter numbers. §8.
- (Q3) Phospho RT: **implement and test both** Chronologer and Prosit iRT; prefer Chronologer when its modification alphabet covers the requested PTMs. §9.
- (Q4) **Agreed** — start on prosit_2020_hcd, do not touch PTMs until Prosit build + apply is proven correct on non-PTM data.

11. Documentation & tracking plan

Per your instruction to keep CLAUDE.md and plan files current “as we go”:

- This directory dev_docs/prosit_ptm_integration/ holds the living docs:
 - PLAN.md / PLAN.pdf — this document.
 - koina_findings.md — the verified API/inference reference (contracts, model list).
 - FINDINGS.md — created in Phase 1 to record benchmark results and decisions (kept updated each phase).
- CLAUDE.md gets a short pointer to this directory once you approve the approach (I did **not** edit CLAUDE.md yet — that would be premature before sign-off).
- Each phase’s completion is recorded in FINDINGS.md with the concrete verification numbers, so the plan and reality stay in sync.

12. Key references

- **Removal commit:** 6bab0dadb (2026-05-01). Reference the pre-removal tree via 6bab0dadb[^] (= a33f96466, 2026-04-23). Cleanest documented multi-model version: 2e289536e (2025-08-29).
- **Working historical Prosit config on disk:** data/test_build_spec_lib/scenario_prosit/params_prosit (uses minimal_protein.fasta, prediction_model="prosit_2020_hcd", max_frag_rank=50).
- **Current code:** src/Routines/BuildSpecLib.jl, src/Routines/BuildSpecLib/{koina, fragments, chronologer}, src/structs/{KoinaStructs, SpectralLibrary}/, src/Pioneer.jl (MODEL_CONFIGS/KOINA_URLS).
- **Verified Koina models:** Prosit_2020_intensity_HCD, Prosit_2024_intensity_PTMs_gl, Prosit_2024_irt_PTMs_gl, Chronologer_RT (PTM-capable). Full list in koina_findings.md.